

Dynamics and Optimization of Rifle Barrel Vibrations

From ballistic theory to finite-element simulation

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1 Introduction: principles of positive compensation

This first section presents the *why* and the *how* of positive compensation without resorting to mathematical tools. A curious reader who only wishes to understand the idea and its practical interest may stop at the end of section 1.8; the subsequent sections then develop the rigorous modelling and the numerical simulation.

1.1 The challenge of precision shooting

In precision-shooting disciplines (for instance, ISSF 50-metre prone), the pursuit of a perfect group runs up against the physical limits of the ammunition. Even with *match-grade* cartridges, there remains an unavoidable variation in muzzle exit velocity — the *muzzle velocity* — from one cartridge to the next. This variation, often measured in feet per second (fps), is typically on the order of a few percent.

1.2 Why a variation in velocity moves the point of impact

A slower bullet spends more time in flight to reach the target and therefore suffers gravitational acceleration for longer: it drops more. Mechanically, on target, this translates into a lower impact than a faster bullet fired with the same setting. If the barrel were a perfectly rigid and stationary tube, the velocity spread of the cartridges would inevitably translate into a *vertical streak* on the target: fast bullets at the top, slow ones at the bottom.

As an order of magnitude, the drop is $h = \frac{1}{2} g (D/\bar{v})^2$ for a distance D and a muzzle velocity $\bar{v}$, so that a velocity spread Δv shifts the impact by $|\Delta h| \approx g D^2 \Delta v / \bar{v}^3$. For a match .22 LR ($\bar{v} \approx 330$ m/s) fired at $D = 50$ m, a spread of 30 fps ($\Delta v \approx 9$ m/s) therefore produces, through time of flight alone, a vertical streak of about 6 mm — easily enough to ruin a match group. It is precisely this spread that positive compensation sets out to cancel.

What this bounds, and why the gain is counter-intuitive. This term is the *only* one a tuned tuner removes: the ammunition's own dispersion, the wind and the shooter are left untouched. Assuming these sources independent, they combine in quadrature, so that a group G can at best shrink to

$$G_{\text{res}} = \sqrt{G^2 - \Delta h^2}, \quad \Delta h \approx g D^2 \Delta v / \bar{v}^3. \quad (1)$$

This bound involves *no* vibratory quantity — neither amplitude, nor frequency, nor tuner mass — and is therefore computed from three figures the shooter already measures: distance, chronograph velocity spread and habitual group size. It is, on that count, the most robust prediction in this document, whereas the absolute values of θ in section 5.5 and beyond remain calibrated orders of magnitude. Its consequence is counter-intuitive: *the better the shooter, the more a tuner returns*. In .22 LR at 50 m with $\Delta v = 10$ m/s, velocity spread accounts for 7.6 mm; on a 20 mm group, removing it gains only 8%, whereas on an 8 mm group the gain reaches 70%. The tuner is thus structurally a *benchrest* tool, which also explains why reports diverge so widely from one shooter to the next. This bound holds only for recoiling firearms, however (cf. section 7).

1.3 The barrel vibrates like a tuning fork

In reality, when the shot is fired, the barrel does not stay still: it begins to vibrate. The powder explosion, the recoil of the firearm, the projectile’s travel through the bore and its friction against the rifling all excite transverse oscillations — that is, up-and-down and side-to-side — that move the muzzle by a few hundredths of a degree, in the manner of a tuning fork ringing after being struck. These vibrations last several tens of milliseconds, well beyond the time it takes for the bullet to travel through the barrel (typically 2 to 3 milliseconds).

During those 2 to 3 milliseconds, the muzzle is therefore in motion: it rises, falls, returns through its equilibrium position, and so on. The exact angle at which the bullet exits depends not only on the shooter’s aim, but also on the *precise instant* of exit.

1.4 What sets the magnitude of the vibrations

Where does this motion come from? The dominant source, emphasised by G. Kolbe [8], is *recoil*: under the thrust of the gases, the whole firearm moves backwards and *rotates about its centre of gravity*. The barrel axis (the bore line, through which the bullet exits) lies *above* that centre of gravity; the recoil thrust therefore acts with a *lever arm* and imparts a couple to the rear of the barrel, exactly as a flick delivered off-centre tips a ruler lying on a table. This couple is the “hammer blow” that makes the tuning fork ring.

Two concrete quantities then govern the *amplitude* of the muzzle motion:

- the **total weight of the firearm**: a heavy rifle absorbs recoil while moving less, and therefore vibrates less;
- the **height of the bore above the centre of gravity**: the larger this lever arm, the larger the couple — and hence the amplitude.

Neither parameter appears in the tuner setting (which acts on the *rhythm* of the vibrations, not on their cause), but together they explain why two rifles with identical barrels but different mountings do not tune the same way. They are reintroduced formally in Section 3.4 as the excitation moment $M_0(t)$.

1.5 The principle of positive compensation

The brilliant idea, proposed as early as 1901 by A. Mallock [9] and formalised for modern shooting by G. Kolbe (Border Barrels) [6, 7], is to *exploit* these vibrations rather than suffer them. If one manages to tune the barrel so that the muzzle is *rising* at the precise moment the bullet leaves it, then:

- a **fast** bullet exits slightly *early*: at that moment, the barrel has not yet had time to rise much, and the bullet is launched at a *lower* angle (figure 1, red trajectory);
- a **slow** bullet exits slightly *late*: the barrel has continued to rise, and the bullet is launched at a *higher* angle (blue trajectory).

The slow bullet is therefore projected *higher* initially; on target, its extra drop (because it spent longer in flight) is exactly compensated by its extra starting height. All bullets, fast or slow, can thus end up at the *same* point of impact. This is what Kolbe calls *positive compensation*.

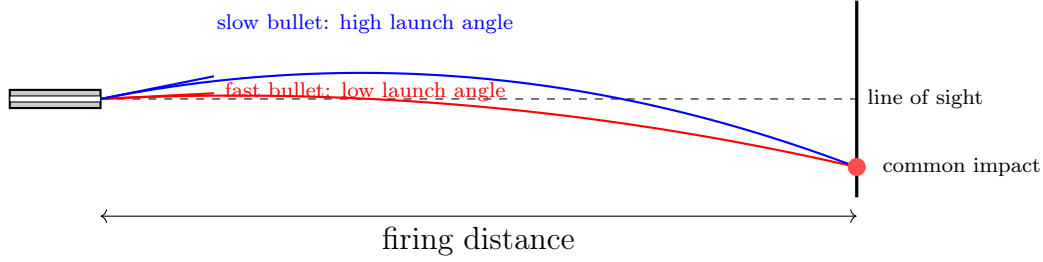


Figure 1: Geometric principle of positive compensation. If the muzzle is rising when the bullets exit, the slower bullets (which leave later) benefit from a higher launch angle. Their extra starting height compensates the extra drop due to their longer flight, and all bullets can end up at the same point of impact on target.

1.6 The role of the tuner

For this phenomenon to occur, one still needs to *match* the barrel’s vibrations to the bullet’s exit time — hence the term *tuner*, which denotes a movable mass (typically 100 to 400 grams) fixed to the muzzle of the barrel. By screwing this mass in or out, the shooter moves it closer to or farther from the muzzle, which subtly modifies the barrel’s vibration frequency. This amounts to adjusting the rhythm of the “tuning fork”: one lengthens or shortens the time it takes for the muzzle to return through its equilibrium position, so that this instant falls precisely when the average bullet exits.

Any change of ammunition or even of lot in principle requires a new adjustment, since the average velocity — and therefore the barrel travel time — changes. Shooters typically proceed by successive trials (the so-called *ladder* or *Audette* method [10]): one fires several series while changing the tuner position by one notch each time, and retains the position for which the groups are tightest.

1.7 Intuitive summary

In summary, positive compensation requires three ingredients:

1. a barrel that vibrates enough to move noticeably between the exit of a fast bullet and that of a slow bullet;
2. a *tuning* of these vibrations that places the exit time of the average bullet at the right point of the oscillation (the *rising* phase);
3. a quantitative compromise between the angular speed of the motion and the ballistic drop: a barrel that rises too slowly will not compensate enough for the dispersion, and a barrel that rises too fast will over-compensate.

The following sections formalise each of these three ingredients and show how the physical parameters of the barrel, the tuner and the ammunition combine to reach the optimum.

1.8 Document outline

The remainder of the document develops the theory and the simulation:

- Section 2: list of notations and conventions used.

- Section 3: mathematical modelling of the barrel (Euler-Bernoulli beam, finite-element method, Newmark- β time integration).
- Section 4: analysis of natural frequencies and sensitivity to the tuner.
- Section 5: mathematical criterion of positive compensation, distinction between temporal node and antinode, experimental confirmation by Kolbe, and validation by the numerical simulation.
- Section 6: practical conclusions for the shooter.

2 Notations and conventions

We adopt the SI system of units (m, kg, s, Pa, rad). The x -axis is aligned with the bore of the barrel, oriented from the breech ($x = 0$) toward the muzzle ($x = L$). The y -axis is transverse, oriented upward. Unless otherwise stated, angles are in radians; the usual conversions are $1 \text{ MOA} = \pi/(180 \times 60) \text{ rad} \approx 2.909 \times 10^{-4} \text{ rad}$ and $1 \text{ mrad} \approx 3.438 \text{ MOA}$.

Symbol	Meaning	Unit
<i>Barrel geometry and material</i>		
L	Barrel length	m
$D_{\text{ext}}, D_{\text{int}}$	Outer (profile) and inner (bore) diameters	m
A	Cross-sectional area $\frac{\pi}{4}(D_{\text{ext}}^2 - D_{\text{int}}^2)$	m ²
I	Second moment of area $\frac{\pi}{64}(D_{\text{ext}}^4 - D_{\text{int}}^4)$	m ⁴
E	Young's modulus (steel ≈ 200 GPa)	Pa
ρ	Mass density (steel ≈ 7850 kg/m ³)	kg/m ³
EI	Flexural rigidity	N·m ²
ρA	Mass per unit length	kg/m
<i>Beam kinematics</i>		
$y(x, t)$	Transverse deflection at point x and time t	m
$\theta(x, t) \equiv \partial y / \partial x$	Slope / angle of the neutral axis	rad
$\theta(L, t)$	Angle at the muzzle (launch angle)	rad
$\dot{\theta} = \partial \theta / \partial t$	Muzzle angular velocity	rad/s
<i>Tuner</i>		
m_t	Tuner mass (typically 100–400 g)	kg
J_t	Tuner transverse moment of inertia	kg·m ²
<i>Natural modes</i>		
ω_n	Angular natural frequency of mode n	rad/s
$f_n = \omega_n / (2\pi)$	Natural frequency of mode n	Hz
$T_n = 1 / f_n$	Natural period of mode n	s
$\phi_n(x)$	Spatial shape of mode n	—
ζ_n	Modal damping ratio	—
<i>Projectile and internal ballistics</i>		
m_p	Projectile mass (.22 LR: ≈ 2.6 g)	kg
$v(x, t)$	Projectile velocity at position x	m/s
v_0, v_{muzzle}	Muzzle velocity (at exit)	m/s
$\bar{v}$	Effective average velocity in the barrel	m/s
t_b	Projectile travel time in the barrel	s
$p(t)$	Pressure in the chamber / bore	Pa
$A_{\text{bore}} = \frac{\pi}{4} D_{\text{int}}^2$	Effective cross-sectional area of the bore	m ²
$\tau_v \equiv -\partial t_b / \partial v_0$	Sensitivity of exit time to muzzle velocity	s/(m/s)
<i>Target and compensation</i>		
D	Shooting distance (50 m for ISSF prone)	m
g	Gravitational acceleration (≈ 9.81 m/s ²)	m/s ²
$h_{\text{bal}}(v_0)$	Ballistic drop at distance D	m
θ_{out}	Muzzle angle at instant t_b : $\theta(L, t_b)$	rad
$\dot{\theta}_{\text{out}}^*$	<i>Target</i> angular velocity for full compensation	rad/s
<i>Numerical simulation</i>		
N	Number of finite elements	—
$L_e = L / N$	Element length	m
$[K], [M], [C]$	Global stiffness, mass, damping matrices ⁶	—
$\{u(t)\}$	Vector of nodal DOFs (translations + rotations)	—
Δt	Newmark- β time step	s

3 Mathematical development

3.1 Continuous model: Euler–Bernoulli beam

We model the barrel as a prismatic beam clamped at the breech ($x = 0$) and free at the muzzle ($x = L$), with cross-sectional area $A(x)$ and second moment of area $I(x)$ (assumed piecewise constant; *bull*, *sporter* and tapered profiles can be handled by varying these element-wise). Writing $y(x, t)$ for the transverse deflection, the partial differential equation governing small oscillations reads:

$$\rho A \frac{\partial^2 y}{\partial t^2} + \frac{\partial^2}{\partial x^2} \left(EI \frac{\partial^2 y}{\partial x^2} \right) = q(x, t), \quad (2)$$

where $q(x, t)$ represents the distributed transverse load. The boundary conditions are:

$$y(0, t) = 0, \quad \frac{\partial y}{\partial x}(0, t) = 0, \quad EI \frac{\partial^2 y}{\partial x^2} \Big|_{x=L} = M_t, \quad \frac{\partial}{\partial x} \left(EI \frac{\partial^2 y}{\partial x^2} \right) \Big|_{x=L} = -F_t,$$

where M_t and F_t respectively denote the moment and the force applied by the tuner (including its inertial effects). For a thick-walled match profile, the Euler–Bernoulli assumption (plane sections remaining plane and normal to the neutral axis) is reasonable as long as $L/D_{\text{ext}} \gtrsim 20$; beyond that, a Timoshenko model accounting for rotational inertia and shear deformation is preferable.

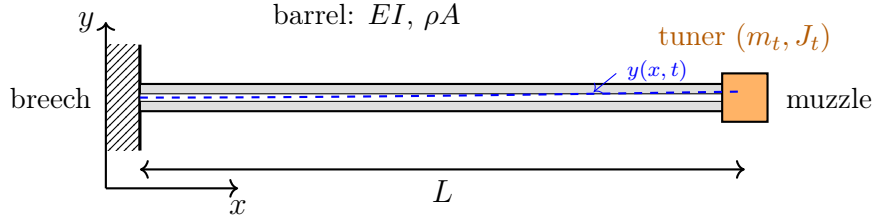


Figure 2: Barrel model: Euler-Bernoulli beam clamped at the breech ($x = 0$), free at the muzzle ($x = L$), carrying at its end a point mass and its rotational inertia (m_t, J_t) representing the tuner. The dashed curve illustrates a transverse deflection $y(x, t)$.

3.2 Finite-element discretisation

We seek a weak solution of (2). Multiplying by a test function $w(x)$ of class $\mathcal{C}^1$ satisfying the essential conditions and integrating by parts twice over $[0, L]$, one obtains the *variational formulation*:

$$\int_0^L \rho A \ddot{y} w \, dx + \int_0^L EI y'' w'' \, dx = \int_0^L q w \, dx + [\text{boundary terms}]. \quad (3)$$

The barrel is partitioned into N elements of length L_e . On each element, two degrees of freedom are assigned per node: deflection y and rotation $\theta = y'$. The local field $y_e(\xi)$, where $\xi \in [0, L_e]$ is the local coordinate, is expressed using **cubic Hermite shape functions**:

$$y_e(\xi) = \mathbf{N}(\xi) \mathbf{u}_e, \quad \mathbf{u}_e = \begin{pmatrix} y_1 \\ \theta_1 \\ y_2 \\ \theta_2 \end{pmatrix}, \quad (4)$$

with

$$\begin{aligned} N_1(\xi) &= 1 - 3\bar{\xi}^2 + 2\bar{\xi}^3, & N_2(\xi) &= L_e (\bar{\xi} - 2\bar{\xi}^2 + \bar{\xi}^3), \\ N_3(\xi) &= 3\bar{\xi}^2 - 2\bar{\xi}^3, & N_4(\xi) &= L_e (-\bar{\xi}^2 + \bar{\xi}^3), \end{aligned}$$

where $\bar{\xi} = \xi/L_e$. These polynomials guarantee the $\mathcal{C}^1$ continuity at the nodes required by the biharmonic operator of (3).

Substituting (4) into (3), one obtains, for each element, the elementary stiffness matrix

$$[K]^e = \int_0^{L_e} EI (\mathbf{N}'')^\top \mathbf{N}'' d\xi = \frac{EI}{L_e^3} \begin{bmatrix} 12 & 6L_e & -12 & 6L_e \\ 6L_e & 4L_e^2 & -6L_e & 2L_e^2 \\ -12 & -6L_e & 12 & -6L_e \\ 6L_e & 2L_e^2 & -6L_e & 4L_e^2 \end{bmatrix}, \quad (5)$$

and the consistent mass matrix

$$[M]^e = \int_0^{L_e} \rho A \mathbf{N}^\top \mathbf{N} d\xi = \frac{\rho A L_e}{420} \begin{bmatrix} 156 & 22L_e & 54 & -13L_e \\ 22L_e & 4L_e^2 & 13L_e & -3L_e^2 \\ 54 & 13L_e & 156 & -22L_e \\ -13L_e & -3L_e^2 & -22L_e & 4L_e^2 \end{bmatrix}. \quad (6)$$

The nodal load vector is obtained analogously as $\mathbf{f}_e = \int_0^{L_e} q(\xi, t) \mathbf{N}^\top d\xi$.

3.3 Assembly, boundary conditions and tuner integration

The global matrices $[K]$ and $[M]$ of size $2(N+1) \times 2(N+1)$ are assembled by summing the elementary contributions on the shared degrees of freedom. The breech clamping imposes $y(0, t) = 0$ and $\theta(0, t) = 0$: we delete the first two rows and columns of the system, which defines the active sub-matrices $[K]_a$ and $[M]_a$.

Adding the tuner at the muzzle (last node, index $N+1$) translates into adding its mass m_t on the translation degree of freedom and its moment of inertia J_t (computed about the axis perpendicular to the barrel axis, in the vibration plane) on the rotation degree of freedom:

$$[M]_a \longleftarrow [M]_a + \underbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{0} & m_t & 0 \\ 0 & 0 & J_t \end{bmatrix}}_{\text{at the muzzle DOFs}}. \quad (7)$$

If the tuner is modelled as a cylinder of mass m_t , radius R_t and length ℓ_t , its moment of inertia about the transverse axis through its centre of gravity is $J_t = m_t (3R_t^2 + \ell_t^2)/12$; an offset d between the tuner's centre of gravity and the muzzle introduces an additional term $m_t d^2$ (parallel-axis theorem) and a mass-rotation coupling $m_t d$ that must be taken into account for a long, offset tuner.

3.4 Modelling of the dynamic excitation

The excitation $\{F(t)\}$ gathers several contributions, heterogeneous in amplitude and spectral bandwidth:

(a) Gas pressure and recoil. The pressure $p(t)$ in the chamber produces a longitudinal force applied at the breech. If the firearm is not held in a perfectly symmetric way (which is always the case), this force induces a transverse moment $M_0(t)$ at the clamped node. We typically model $p(t)$ as a Pierret profile or a Gaussian fit to measured pressure–time curves, with peak amplitude $p_{\max} \sim 200$ MPa for the .22 LR and characteristic duration ~ 0.5 ms.

Physically, this moment is not an artefact of an asymmetric hold: it exists even for an ideally held firearm, because the *bore line is offset from the centre of gravity* of the rifle. Following Kolbe’s modelling [8], the free firearm of mass m_r recoils and rotates about its centre of gravity, located a distance h_{cg} below the bore line; the recoil force $F_{\text{rec}}(t) = p(t) A_{\text{bore}}$ thus acts with this lever arm and imparts to the base of the barrel the moment

$$M_0(t) = p(t) A_{\text{bore}} h_{\text{cg}}, \quad (8)$$

while the resulting angular amplitude decreases as $1/m_r$ (a heavy rifle recoils less). The clamped model does not contain the rigid-body recoil motion; we therefore tie the lever arm h_{offset} of Section 5.5 to the physical recoil moment by factoring its dependence,

$$h_{\text{offset}}(m_r, h_{\text{cg}}) = h_{\text{offset}}^{\text{ref}} \frac{h_{\text{cg}}}{h_{\text{cg}}^{\text{ref}}} \frac{m_r^{\text{ref}}}{m_r}, \quad (9)$$

where the *single* constant $h_{\text{offset}}^{\text{ref}}$ is fixed once and for all for the reference rifle ($m_r^{\text{ref}} = 5$ kg, $h_{\text{cg}}^{\text{ref}} = 25.4$ mm — the default values of Kolbe’s simulator, not a measurement), and bounded by Vaughn’s [5] excitation measurement (section 5.5). Rather than recalibrating for each rifle, one then *predicts* the dependence on two quantities measurable on the actual rifle — the total weight m_r (easily weighed) and the bore-to-centre-of-gravity distance h_{cg} (trickier, obtained by balancing): mounting the same barrel on a lighter rifle, or one whose bore sits higher above the centre of gravity, amplifies the vibrations and calls for a fresh tuner setting.

(a’) Boundary conditions: clamping vs a compliant action. The model above assumes a perfect clamp at the breech ($y(0) = 0$, $\theta(0) = 0$). This is an idealisation: Kolbe [8] prefers *not* to fix the base and represents the compliance of the receiver by an additional beam segment (~ 100 mm long, ~ 38 mm diameter, ~ 25 mm bore) appended at the rear, the whole assembly recoiling *freely* in space (the *free-recoil* assumption, well approximated by bag shooting but violated if a light rifle is firmly shouldered). Our rigid clamp therefore slightly overestimates the fundamental frequency and ignores the rigid-body recoil motion; it nonetheless remains relevant for the *relative* muzzle angle $\theta(L, t)$, the only quantity that drives compensation (cf. Section 5), and it simplifies the modal analysis. The “compliant action” refinement is the natural extension for quantitative work on a real rifle.

(b) Moving projectile load (weight effect). The projectile, of mass m_p , applies at its current position $x_p(t)$ a transverse force due to gravity (its weight) and to the engagement with the rifling. To first approximation, the projectile is treated as a moving point force:

$$q_p(x, t) = -m_p g \delta(x - x_p(t)), \quad (10)$$

where g is the gravitational acceleration and δ the Dirac distribution. The corresponding nodal vector is obtained by evaluating the shape functions at position $x_p(t)$.

(c) **Gyroscopic torque of the rifling.** The angular acceleration imparted by the rifling transmits a reactive torque to the barrel, which can be modelled as a distributed moment proportional to $\dot{v}(x_p)$. This effect, weaker than (a) and (b), can be neglected in a first analysis.

(d) **Projectile position.** The internal kinematics $x_p(t)$ is obtained by integrating the internal-flow equation:

$$m_p \ddot{x}_p = p(t) A_b - F_{\text{frict}}(x_p, \dot{x}_p), \quad (11)$$

where A_b is the effective bore cross-section and F_{frict} the engagement resistance. The exit time t_b satisfies $x_p(t_b) = L$; for .22 LR Match in a 26" barrel, $t_b \approx 1.5$ to 1.7 ms.

3.5 Newmark- β time-integration scheme

The semi-discrete equation of the structure reads

$$[M]\{\ddot{u}(t)\} + [C]\{\dot{u}(t)\} + [K]\{u(t)\} = \{F(t)\}, \quad (12)$$

where $[C]$ is usually modelled by Rayleigh damping $[C] = \alpha_M[M] + \alpha_K[K]$ with two constants (α_M, α_K) calibrated against measured modal damping ratios ($\zeta_1 \sim 0.5\%$ to 1% for a steel barrel).

The implicit Newmark- β scheme advances from step n to step $n+1$ (with Δt) by:

$$\{u_{n+1}\} = \{u_n\} + \Delta t \{\dot{u}_n\} + \frac{\Delta t^2}{2}((1-2\beta)\{\ddot{u}_n\} + 2\beta\{\ddot{u}_{n+1}\}), \quad (13)$$

$$\{\dot{u}_{n+1}\} = \{\dot{u}_n\} + \Delta t((1-\gamma)\{\ddot{u}_n\} + \gamma\{\ddot{u}_{n+1}\}), \quad (14)$$

where $\{\ddot{u}_{n+1}\}$ is obtained by re-injecting (13) into (12) evaluated at t_{n+1} , leading to the linear system:

$$\left([M] + \gamma \Delta t [C] + \beta \Delta t^2 [K]\right) \{\ddot{u}_{n+1}\} = \{F_{n+1}\} - [C]\{\dot{u}_n^*\} - [K]\{u_n^*\}, \quad (15)$$

where $\{u_n^*\}$ and $\{\dot{u}_n^*\}$ are the predictors built from the current values. The classic pair $(\gamma, \beta) = (1/2, 1/4)$ (*average constant acceleration*) is *unconditionally stable* and conserves energy with no numerical dissipation; it is favoured here to faithfully preserve the amplitude of excited modes. The time step must nevertheless resolve the highest frequency of interest: we take $\Delta t \lesssim T_{\text{max}}/20$, where T_{max} is the period of the highest physically significant mode (typically a few kHz for a barrel, hence $\Delta t \sim 10 \mu\text{s}$).

4 Modal analysis and sensitivity to the tuner

4.1 Eigenvalue problem

In the absence of damping and excitation, we seek harmonic solutions $\{u(t)\} = \{\phi\} e^{i\omega t}$, which reduces (12) to a generalised eigenvalue problem:

$$([K] - \omega^2[M]) \{\phi\} = \mathbf{0}. \quad (16)$$

The solutions $(\omega_n^2, \{\phi_n\})$ give the natural angular frequencies ω_n and the mode shapes $\{\phi_n\}$. The frequencies in hertz are $f_n = \omega_n/(2\pi)$. For the reference barrel of the simulator (steel, $L = 0.66$ m, $D_{\text{ext}} = 24$ mm cylindrical, $D_{\text{int}} = 5.6$ mm), a 20-element FEM solution gives the following first modes:

Mode	Bare barrel	200 g tuner	400 g tuner
f_1 (Hz)	39.87	34.16	30.33
f_2 (Hz)	230.1	217.66	206.9
f_3 (Hz)	633.5	602.39	575.1
f_4 (Hz)	≈ 1.19 kHz	1.14 kHz	1.09 kHz
f_5 (Hz)	≈ 1.87 kHz	1.79 kHz	1.72 kHz

The analytical solution for the bare cantilever beam, $f_1 = \frac{1.875^2}{2\pi} \sqrt{EI/(\rho AL^4)}$, gives $f_1 \approx 40.0$ Hz, in perfect agreement with the FEM result (39.87 Hz). Adding a 200 g tuner at the muzzle drops f_1 by about 14 % (from 39.9 to 34.2 Hz); a 400 g tuner gives a 24 % reduction. This range largely covers the useful time window around $t_b \approx 2$ –3 ms.

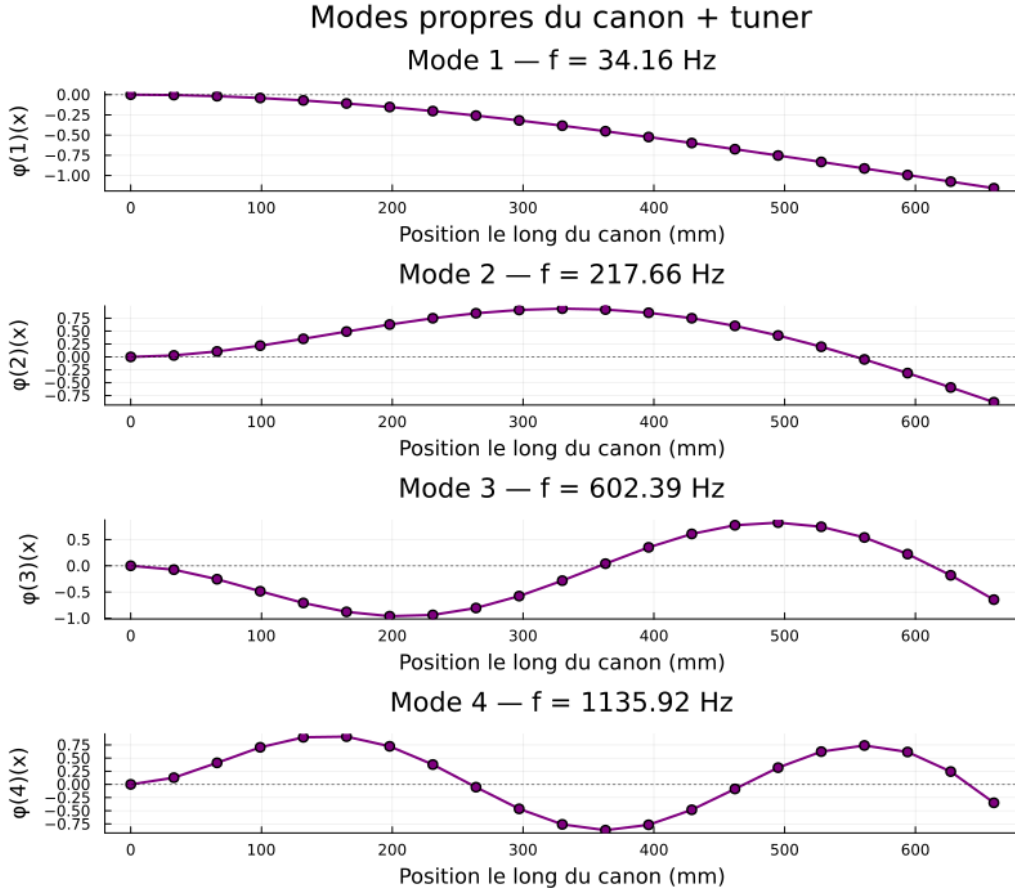


Figure 3: First four mode shapes $\phi_n(x)$ of the barrel fitted with a 200 g tuner (FEM calculation, 20 elements). Note the monotonic shape of mode 1 (spatial antinode at the muzzle), and the progressive migration of the spatial nodes of higher modes toward the muzzle, a consequence of mass concentration at $x = L$.

4.2 Sensitivity via the Rayleigh quotient

The *Rayleigh quotient* associated with mode n reads

$$\omega_n^2 = \frac{\{\phi_n\}^\top [K] \{\phi_n\}}{\{\phi_n\}^\top [M] \{\phi_n\}}. \quad (17)$$

Let $\phi_n^{(L)}$ denote the mode component at the muzzle node (in translation). A perturbation δm_t of the tuner mass modifies $[M]$ by $\delta m_t \phi_n^{(L)2}$ in the denominator, without affecting $[K]$. To first order, we obtain:

$$\boxed{\frac{d\omega_n^2}{dm_t} \approx -\omega_n^2 \frac{|\phi_n^{(L)}|^2}{\{\phi_n\}^\top [M] \{\phi_n\}} \leq 0.} \quad (18)$$

Two important conclusions follow:

1. Adding mass at the muzzle *always lowers* the frequency (negative sign).
2. The reduction is more pronounced when the mode has a large amplitude at the muzzle. For the fundamental mode of a cantilever beam, $\phi_1^{(L)}$ is maximum, so f_1 drops strongly; for mode 2, the vibrational node moves closer to the muzzle, and the tuner's effect is more modest.

These properties form the basis of the tuning mechanism: by moving the position of a sliding mass (or stacking weights), the shooter continuously sweeps f_1 over a typical range of ± 15 to ± 30 %.

4.3 Modal decomposition and transient response

By decomposing the response onto the basis of mass-normalised mode shapes, $\{u(t)\} = \sum_n q_n(t) \{\phi_n\}$, equation (12) decouples into

$$\ddot{q}_n(t) + 2\zeta_n \omega_n \dot{q}_n(t) + \omega_n^2 q_n(t) = f_n(t), \quad f_n(t) = \{\phi_n\}^\top \{F(t)\}. \quad (19)$$

For an impulsive excitation (the pressure+projectile signature resembles a pulse of a few hundred microseconds), each mode begins a free oscillatory regime. The global rotational response at the muzzle is then

$$\theta(L, t) = \sum_n \phi_n^{\theta(L)} q_n(t), \quad (20)$$

where $\phi_n^{\theta(L)}$ is the rotation component of mode n at the muzzle node. In practice, the first mode dominates the muzzle kinematics at characteristic times $\sim t_b$; it is therefore essentially its tuning that the tuner targets.

A transient regime, not an established standing wave. A caveat is in order, one that Kolbe [8] makes the heart of his argument: it is tempting to represent “how a barrel vibrates” by the *analytic standing-wave solutions* of the beam equation. Yet these waves do *not* form during the useful window. Their phase velocity is too low for a steady standing-wave regime to establish itself over the $t_b \approx 1\text{--}3$ ms of bullet passage: at that instant the barrel is still within the *transient response* to the impulsive recoil moment, not in a settled periodic regime. This is precisely why the modal decomposition above is exploited here as a *forced transient response* (integrated step by step by Newmark- β) rather than as a superposition of modes with frozen amplitudes, and why the reasoning of Section 5.3 favours the *temporal node* (the instantaneous state of the oscillation at t_b) over the spatial node (a property of a fully established mode).

5 Muzzle kinematics and positive compensation

5.1 Travel time and muzzle angle

Let $v(x, t)$ denote the projectile velocity at position x ; the exit time is

$$t_b = \int_0^L \frac{dx}{v(x)}. \quad (21)$$

The effective launch angle at exit reads

$$\theta_{\text{out}} = \theta(L, t_b) = \left. \frac{\partial y}{\partial x} \right|_{x=L, t=t_b}. \quad (22)$$

A variation $\Delta\theta_{\text{out}}$ produces a vertical displacement on target (at distance D , to first order without ballistic correction):

$$\Delta h \approx D \Delta\theta_{\text{out}}. \quad (23)$$

The underlying geometry is that of figure 1 in the introduction: what section 1 described qualitatively is here reformulated in terms of muzzle angle and on-target effect.

5.2 Coupling between velocity spread and vertical dispersion

A variation Δv_0 of the muzzle velocity induces two cumulative effects:

1. *Additional ballistic drop.* At distance D , for a flat trajectory, the drop below line of sight is $h_{\text{bal}}(v_0) \simeq gD^2/(2v_0^2)$, hence

$$\frac{\partial h_{\text{bal}}}{\partial v_0} \simeq -\frac{gD^2}{v_0^3} < 0. \quad (24)$$

Numerically, at $D = 50$ m and $v_0 \approx 318$ m/s (1043 ft/s, Eley Tenex), $\partial h_{\text{bal}}/\partial v_0 \approx -0.76$ mm/(m/s), i.e. -0.016 MOA per ft/s: exactly the number computed by Kolbe [6] using a trajectory solver. The choice of v_0 matters and must remain *consistent* with the ammunition on which τ_v was measured (Eley Tenex, see below): at $v_0 = 308$ m/s one would obtain -0.0176 MOA per ft/s, 10 % too high, and the error would propagate all the way to criterion (28).

2. *Exit-time shift.* The internal kinematics couples t_b and v_0 : in the limit, $t_b \propto 1/v_0$ gives $\partial t_b/\partial v_0 = -t_b/v_0$, but the actual dependence is stronger because a more energetic charge raises the pressure *throughout* the bore and shortens t_b more than a simple kinematic ratio. We therefore write

$$\tau_v \equiv -\frac{\partial t_b}{\partial v_0} > 0, \quad (25)$$

a quantity directly measurable using a chronograph coupled to a muzzle-exit sensor. Kolbe reports [6], for Eley Tenex in a 26" barrel,

$$\tau_v \approx \frac{1 \text{ ms}}{375 \text{ ft/s}} = \frac{1 \text{ ms}}{114 \text{ m/s}} \approx 8.8 \text{ } \mu\text{s} / (\text{m/s}), \quad (26)$$

and argues that this value is, to within $\pm 10\%$, a constant for rimfire barrels longer than 6 inches.

The positive-compensation condition requires the two contributions to cancel on target:

$$\boxed{\left. \frac{d\theta_{\text{out}}}{dt} \right|_{t_b} \cdot \Delta t_b + \frac{1}{D} \frac{\partial h_{\text{bal}}}{\partial v_0} \Delta v_0 = 0.} \quad (27)$$

Substituting $\Delta t_b = -\tau_v \Delta v_0$ and (24), we obtain the *optimal angular velocity* at the muzzle at the moment of exit:

$$\dot{\theta}_{\text{out}}^* = \frac{1}{D \tau_v} \frac{\partial h_{\text{bal}}}{\partial v_0} = -\frac{g D}{v_0^3 \tau_v}. \quad (28)$$

The negative signs of $\partial h_{\text{bal}}/\partial v_0$ and $\partial t_b/\partial v_0$ combine to make $\dot{\theta}_{\text{out}}^*$ positive: the muzzle must therefore be rising ($\dot{\theta} > 0$) at the moment of shot release. Numerically, with $D = 50$ m, $v_0 = 318$ m/s and $\tau_v = 8.8 \mu\text{s}/(\text{m/s})$:

$$\dot{\theta}_{\text{out}}^* \approx 1.73 \text{ mrad/ms} \approx 5.96 \text{ MOA/ms},$$

to be compared with the 6.0 MOA/ms experimentally measured by Kolbe: *the agreement is better than 1%*. This is not fortuitous — the two routes are independent. Kolbe multiplies his measured natural drop (0.016 MOA per ft/s) by his exit-time sensitivity (375 ft/s per ms); we start from the ballistic kinematics (24) and τ_v . That the two converge to within 1% validates the chain of reasoning — provided v_0 is kept consistent with the reference ammunition.

5.3 Spatial node vs temporal node: a semantic clarification

The vocabulary of *node* and *antinode*, inherited from the acoustics of standing waves, often causes confusion when applied to shooting. Two physically distinct meanings coexist:

Spatial node/antinode — special points of the *mode shape* $\phi_n(x)$ along the barrel: a *spatial node* is an x^* such that $\phi_n(x^*) = 0$ (zero transverse displacement); a *spatial antinode* is an x^* where $|\phi_n|$ reaches a local maximum. For the fundamental mode of a cantilever beam, the shape is monotonically increasing from 0 (clamped end) to $\phi_1(L)$ (muzzle). **The muzzle is therefore, by construction, a spatial antinode of mode 1.** No choice of tuner moves this property: the muzzle remains where the bullet exits, at $x = L$.

Higher modes ($n \geq 2$), on the other hand, exhibit spatial nodes *inside* the barrel. Placing the tuner precisely at a spatial node of mode n makes that mode insensitive to the added mass (cf. generalisation of (18) to a mass positioned away from $x = L$). This is the trick of so-called *harmonically-balanced* tuners: f_1 can be adjusted without disturbing f_2 or f_3 , which makes the adjustment more monotonic and reproducible.

Temporal node/antinode — special points of the *in-time* oscillation of the muzzle angle $\theta(L, t)$: a *temporal node* is an instant t^* where $\theta(L, t^*) = 0$ (zero crossing); a *temporal antinode* is an instant where $|\theta(L, t)|$ reaches a maximum (i.e. $\dot{\theta}(L, t^*) = 0$). For a nearly sinusoidal mode 1, these two types of instants alternate at a quarter period.

The criterion (28) then translates unambiguously:

Exit instant t_b	$\theta(L, t_b)$	$\dot{\theta}(L, t_b)$
Temporal antinode	$\pm\Theta_1$ (max)	0
Temporal node	0	$\pm\Theta_1 \omega_1$ (max)

Positive compensation requires exit in the vicinity of an *ascending temporal node*, where $\theta = 0$ and $\dot{\theta} > 0$ is at its maximum. Exiting at the peak of the oscillation (temporal antinode) is, on the contrary, the *worst* case: not only does $\dot{\theta} = 0$ wipe out any compensation, but one also fires with a maximum static angular offset that shifts the mean point of impact.

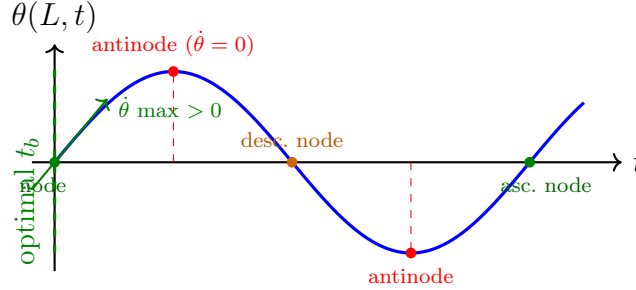


Figure 4: Temporal node vs antinode distinction on the muzzle angle $\theta(L, t)$. Positive compensation requires t_b to fall in the vicinity of an *ascending temporal node* (green point at $t = 0$ and $t = T_1$), where $\theta = 0$ but $\dot{\theta}$ is maximum and positive. Exiting at a temporal antinode (red points, $\dot{\theta} = 0$) cancels the compensation.

What Kolbe [6, 7] calls the “upward swing of the vibration at the muzzle” corresponds precisely to this *ascending temporal node*. In the competitor jargon (*ladder* or *OCW* [10] method), the “node” of tuning refers more loosely to a zone of *robustness* of grouping, i.e. a plateau around which a small shift of t_b does not degrade the group. In the strict sense of vibration mechanics, this plateau is precisely the window around an ascending temporal node — where $\ddot{\theta}(L, t_b)$ is close to zero and $\dot{\theta}(L, t_b)$ hovers near its extremal value, which makes the tuning weakly sensitive to fluctuations of t_b .

5.4 Experimental confirmation (Kolbe, 2015)

Kolbe [7] directly verified the criterion (28) on an instrumented test bench measuring the muzzle angle optically (crossed polariser) with a photo-detector gate locating the exact exit instant. On a 26" Border barrel, .22 LR calibre, Eley EPS Tenex ammunition, two configurations are compared:

Configuration	measured $\dot{\theta}(L, t_b)$	On-target groups
Bare barrel	−9.4 MOA/ms (muzzle dropping)	pronounced <i>vertical</i> string
Barrel + 200 g tuner	+6.0 MOA/ms (muzzle rising)	<i>round</i> groups

The value +6.0 MOA/ms coincides with criterion (28) for 50 m and the ammunition used. The near-complete elimination of vertical stringing experimentally confirms that the criterion is not only necessary (sign) but also sufficient (magnitude) when higher-order modes remain weakly excited.

In particular, Kolbe notes that the *vertical translation velocity* $\dot{y}(L, t_b)$ of the muzzle contributes negligibly to the on-target dispersion compared to $\dot{\theta}(L, t_b)$: it is indeed the muzzle’s *rotation*, not its translation, that determines the effective launch angle. This observation justifies *a posteriori* the DOF split (translation + rotation) of the FEM model, and orients the control toward extracting the rotational component at the terminal node.

5.5 Validation by FEM + Newmark- β simulation

The `Julia` simulator accompanying this document (`simulation.jl`) implements the full formalism: FEM assembly, clamping, tuner, internal ballistics (“burnout” profile: acceleration over a fraction $\phi = 0.35$ of the barrel, then coast, calibrated on Kolbe’s τ_v), excitation by the breech recoil moment $M(t) = p(t) A_{\text{bore}} h_{\text{offset}}$, moving projectile load, Rayleigh damping and Newmark- β scheme. There the lever arm h_{offset} is 16.5 mm. It is an *effective* quantity, not a dimension: it places the model at the compensation optimum for a tuner setting lying within the real adjustment range. Its physical referent comes from the only published measurement of the excitation itself, Vaughn’s [5] (strain gauges on the forward receiver ring of a .270 Win, chapter 4): the *applied* moment he reports, ~ 1500 in-lb, corresponds to a lever arm of 11.9 mm, hence ~ 9 to 12 mm once carried over to .22 LR through (9). The ratio, of order 1.4 to 1.8, *quantifies* what the clamped frame omits — the rigid-body rotation of the firearm under recoil. Adopting the physical value instead, in the amputated model, makes it unusable: it then saturates at 3.5 MOA/ms and never reaches the 6.0 optimum, in mass or in position. The only quantity confronted with measurement remains $\dot{\theta}(L, t_b)$.

The internal-ballistics profile now reproduces τ_v . The “burnout” profile (constant acceleration over a fraction $\phi = 0.35$ of the barrel, then coast) realises $\tau_v = -\partial t_b / \partial v_0 = (1 + \phi) L / v^2 = 8.8 \mu\text{s}/(\text{m/s})$, exactly the value measured by Kolbe and used in criterion (28). The former exponential lag $v = v_m(1 - e^{-t/\tau})$ was *front-loaded* (all acceleration at the breech, then coast): its sensitivity capped at $L/v^2 \approx 6.5 \mu\text{s}$, 35% too low, the bullet exiting too early ($t_b = 2.46$ vs 2.80 ms). Kolbe’s kinematic chain — natural drop, exit-time sensitivity, required rate — is thus *fully* reproduced (`kolbe_validation.jl`: $0.016 \times 375 = 6.0$ MOA/ms recovered to 1%).

What remains out of reach is the *absolute amplitude*. At the correct exit time $t_b = 2.80$ ms, the clamped model gives $\dot{\theta}(L, t_b)$ of the right *sign* — negative for the bare barrel (≈ -1.1 MOA/ms, muzzle descending, like Kolbe’s measured -9.4), brought toward positive by the muzzle mass (+2.6 at 400 g) — but about three times too small to reach the measured 6.0 MOA/ms without an implausible vibration amplitude. This is the symptom, already encountered, of the *rigid-body rotation* absent from the clamped model, not a defect of the ballistic profile. Notably, correcting the profile *improves* the qualitative agreement: at the former $t_b = 2.5$ ms the bare barrel came out *rising* (wrong sign); at the correct time it comes out *descending*, consistent with Kolbe.

This amplitude deficit has since been *lifted* by the free-recoiling rifle model (`haral_rifle_sweep.jl` `kolbe_amplitude.jl`): with the physically anchored recoil moment ($\int A_{\text{bore}} p dt = (1 + \beta) m_p v$), the peak of $|\dot{\theta}|$ reaches ~ 5 MOA/ms for a one-inch bore height, ~ 10 for two inches — Kolbe’s order of magnitude, with no fitting parameter. Yet the rigid-body rotation contributes only ~ 0.2 MOA/ms: it is not what supplies the amplitude, but the elastic *whip* it excites once the breech is freed. Only the *exact* match of $\dot{\theta}(L, t_b)$ — its phase at exit — remains out of reach, depending on the real whip frequency and exit time, which are unpublished.

These plots illustrate a distinction essential for the shooter: the tuner acts *primarily on the mean point of impact* (via $\theta(L, t_b)$) and *secondarily on the vertical dispersion* (via $\dot{\theta}(L, t_b)$). When $t_b \ll T_1$, the optimality window in $\dot{\theta}$ is wide and flat; the tuner adjustment then mainly serves to compensate for the velocity spread, while the sight adjustment takes care of the POI shift.

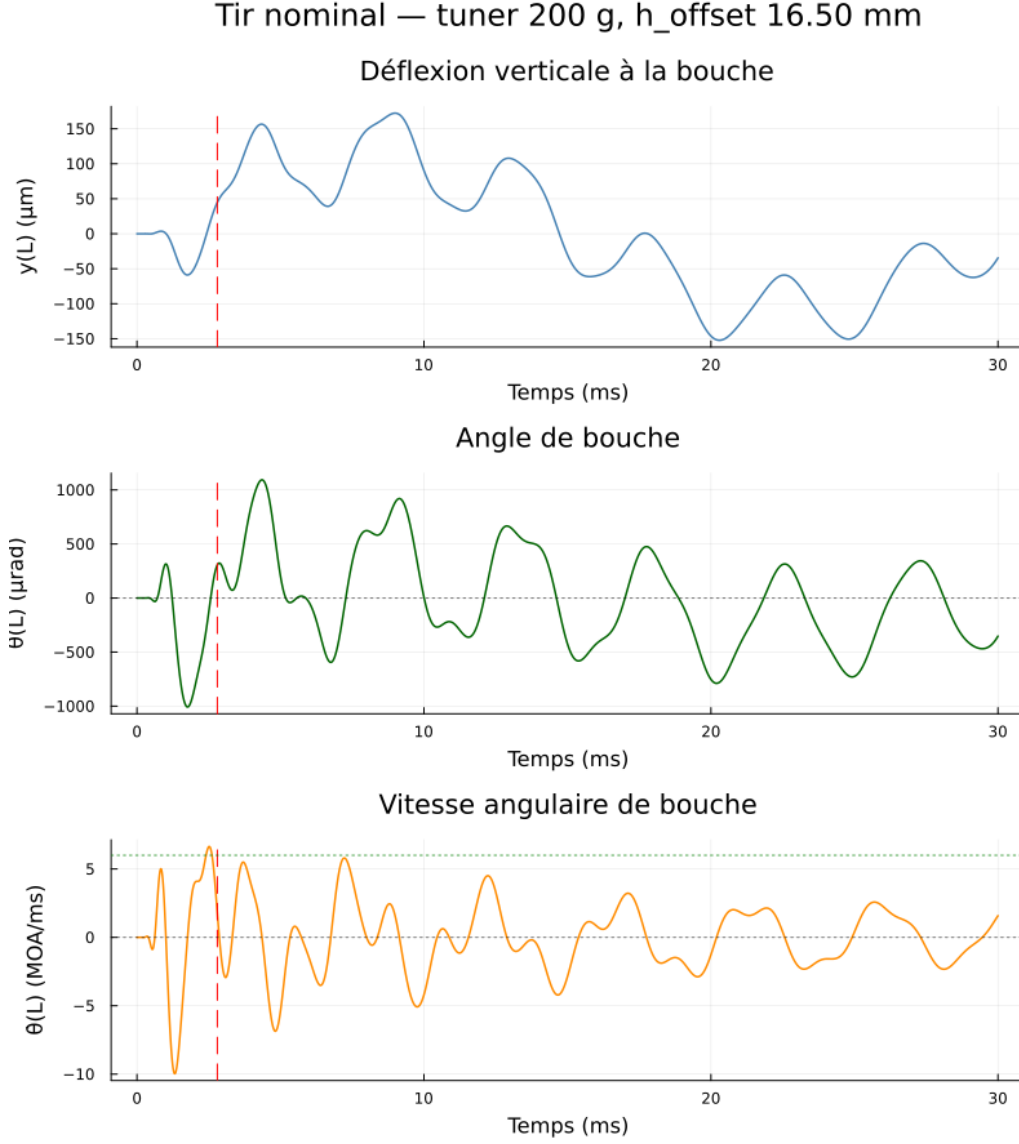


Figure 5: Simulated transient response for the nominal configuration (200 g tuner, $h_{\text{offset}} = 16.5$ mm). From top to bottom: deflection $y(L, t)$, muzzle angle $\theta(L, t)$, and angular velocity $\dot{\theta}(L, t)$. The red dashed line marks $t_b = 2.80$ ms (burnout profile); the green dotted line indicates the Kolbe target (6 MOA/ms). At t_b the model gives $\dot{\theta} = +1.8$ MOA/ms: of the right sign (rising, tuner fitted) but of amplitude below the target — see the caveat below.

5.6 Tuning by position: the effective adjustment parameter

The sweep in figure 6 varies the *mass* m_t , a convenient parameter in the model. It is not, however, the adjustment variable used in the field: there the mass is *fixed* (a given weight is fitted) and tuning is done by *screwing* the tuner further out or in, overhanging ahead of the muzzle — a continuous adjustment counted in thread turns.

This degree of freedom is modelled by offsetting the tuner mass a distance d ahead of the muzzle node. The mass–rotation coupling of an offset mass adds to the degree-of-

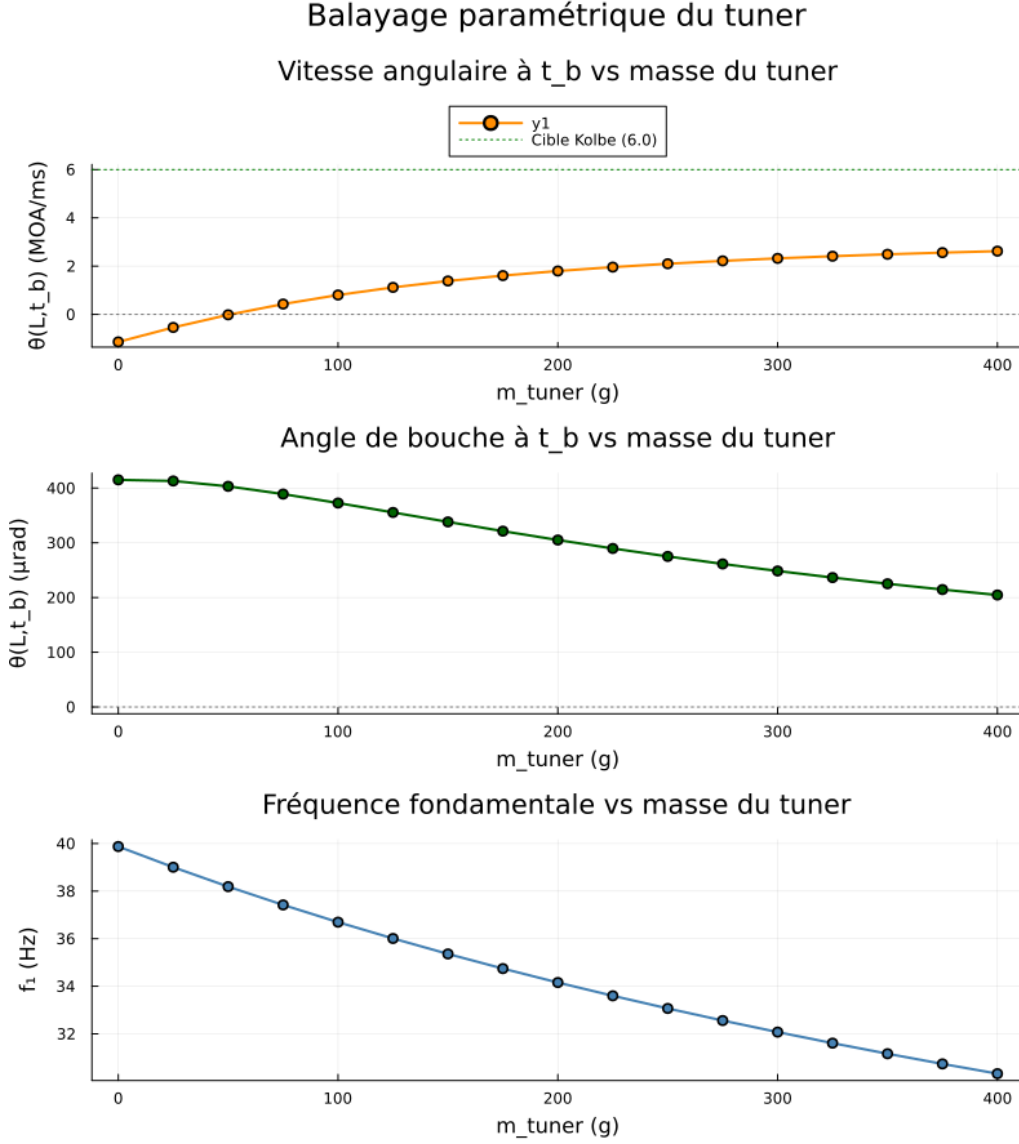


Figure 6: Parameter sweep on the tuner mass $m_t \in [0, 400]$ g (other parameters fixed). From top to bottom: (i) $\dot{\theta}(L, t_b)$ changes sign — -1.1 MOA/ms for the bare barrel (muzzle descending at exit) toward $+2.6$ at 400 g (rising): this is Kolbe’s mechanism, the tuner taking the muzzle from descending to rising; (ii) $\theta(L, t_b)$ decreases monotonically from $+416$ to $+205$ μrad , reflecting a shift of the mean point of impact; (iii) f_1 decreases monotonically from 39.9 to 30.3 Hz (relation predicted by the Rayleigh quotient, equation (18)).

freedom pair (y_L, θ_L) the contribution

$$\mathbf{M}_{\text{add}} = \begin{bmatrix} m_t & m_t d \\ m_t d & m_t d^2 + J_{\text{cm}} \end{bmatrix}, \quad (29)$$

where J_{cm} is the tuner’s own inertia about its centre of mass; at $d = 0$ the point mass of the previous model is recovered.

Three results emerge. First, *position alone suffices to tune*: there is no need to change weights, as the tuner travel spans the whole useful range. Second, *the weight determines where along the travel*, not whether tuning is possible: the target is reached near $d \approx$

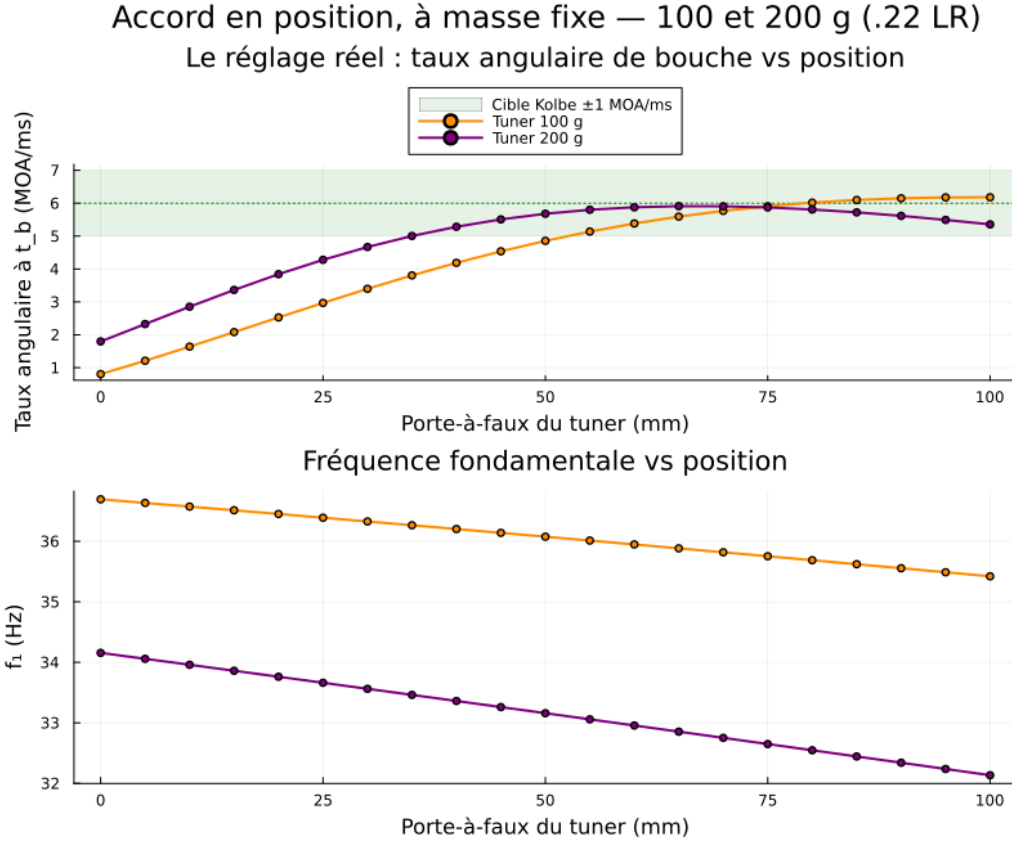


Figure 7: Tuning by position at fixed mass, for two masses bracketing the real range (100 and 200 g). Top: $\theta(L, t_b)$ against the overhang d ; the green band marks ± 1 MOA/ms around Kolbe’s target. The overhang alone moves the angular rate from $+0.8$ to $+6.2$ MOA/ms (100 g) or from $+1.8$ to $+5.9$ (200 g): in both cases the travel spans the entire gap between an uncompensated rifle and the target. Bottom: f_1 decreases with d , position shifting the same modal frequencies as mass does.

80 mm at 100 g and near $d \approx 65$ mm at 200 g — the lighter the mass, the further the sweet spot from the muzzle. Third, *the optimum is broad*, and the more so the heavier the mass: the region within 1 MOA/ms of the target spans ~ 45 mm of travel at 100 g and ~ 65 mm at 200 g. Mass and position thus form a single *tuning space*, the weight fixing the curve and the position tuning along it.

Typical masses. Manufacturers’ published weights bracket the range:

Tuner	Published weight	Intended use
PMA 7/8-32	4.2 oz (119 g)	centrefire
Ezell PDT <i>light</i>	5 oz (142 g)	centrefire
Ezell PDT standard	7 oz (198 g)	centrefire
Harrell’s Rimfire/Air	8 oz (227 g)	rimfire
Starik/Centra carbon tube	~ 200 to 220 g (complete assembly)	rimfire ISSF

that is, ~ 115 to 230 g for a muzzle tuner. These figures do not, however, describe comparable objects: with Harrell’s or Ezell the entire body constitutes the mass, tuned by screwing it over a few millimetres of thread; the Starik carbon tube is the opposite

architecture, a light tube in which a small mass slides over several centimetres, so that its ~ 200 g assembly weight is not the moving mass, which is not published. The two curves in figure 7 bracket both regimes.

Aim for the node, not the number. Section 5 states the compensation condition in two equivalent forms: an angular rate of 6 MOA/ms, but also and above all an *exit near a rising temporal node*, where $\theta = 0$ and $\dot{\theta}$ is maximal. The two forms coincide by quadrature. Tuning on the *number*, however, does not always lead there, and the sweep above supplies a counter-example. As long as the excitation was assumed identical on every shot, the discrepancy had no visible consequence; once it varies (point 12) it is paid for, the sensitivity to that variability being proportional to the *absolute* angle $\theta(t_b)$. At 200 g the model saturates at 5.91 MOA/ms and never reaches 6.0: “closest to target” therefore falls back on the maximum, i.e. on the node, and the 65 mm setting is already the right one. At 100 g, by contrast, $\dot{\theta}$ crosses 6.0 as early as 80 mm and keeps rising to 6.18: aiming at the nominal value stops *short* of the maximum, 131 μ rad away from neutral instead of 16. The variability simulator (`variability.jl`) puts the gap at a **factor of 3** in predicted dispersion (0.70 against 0.23 mm standard deviation at 50 m), the optimum moving to 105 mm. The 80 mm quoted above is therefore a *methodological artefact*, not an optimum. The result also vindicates the *ladder tune*: a shooter who sweeps the travel and keeps the best group finds the node without computing it, where aiming at an angular-rate figure may lead away from it. *Status*: the mis-placement follows from the criterion of section 5 and is readable off the sweep; its quantified *cost*, however, depends on the variability model and its least secure parameter — a testable, unvalidated prediction.

Fineness of adjustment. The steps recorded on a Starik/Centra tuner — weight carrier in 10 mm increments, fine adjustment at 5 turns per 2.5 mm, i.e. a 0.5 mm pitch — correspond, on the curve of figure 7, to ~ 0.01 MOA/ms per turn near the optimum (up to ~ 0.04 on the steep part). The width of the optimum, ~ 90 turns, explains why a *ladder tune* converges without demanding turn-level precision.

6 Practical conclusions for the shooter

The preceding formalism, validated numerically by the finite-element simulation, leads to several exploitable conclusions.

1. Effect of a muzzle mass. Adding a mass at the muzzle *lowers* the fundamental frequency (cf. (18)). For the reference barrel, the adjustment range of a 100–400 g tuner covers an f_1 variation of the order of 8 to 24% (39.9 Hz $\rightarrow$ 30.3 Hz). The equivalent temporal shift in the vibration cycle is of the order of a fraction of a millisecond, which suffices to sweep the entire optimality window in which $t_b \approx 2\text{--}3$ ms must be positioned.

2. Direction of adjustment. Extending the effective tuner position (screwing it outward) increases J_t and the lever arm, which amplifies the mass effect and lowers f_1 . This is the action that “slows down” the muzzle kinematics and delays the zero crossing of $\theta(L, t)$: useful for slower ammunition (delayed exit t_b).

3. Sensitivity to ammunition. Any change of lot or brand modifies $\bar{v}$ and therefore t_b . The optimality window ($\dot{\theta}_{\text{out}}$ near θ_{out}^*) is narrow: a 0.1 ms shift on t_b may represent a non-negligible fraction of the period $T_1 = 2\pi/\omega_1$. Any new lot must therefore be the subject of a systematic retuning at the tuner notch (*ladder tune* or *Audette* method).

4. Physical limits. The tuner does not remove the intrinsic dispersion of the ammunition: it converts a vertical dispersion into a genuinely tighter group only if (27) is satisfied to within a few MOA/ms. The horizontal dispersion, independent of the $(\Delta v_0, \theta_{\text{out}})$ pair, is not compensated. The expected on-target gain therefore remains bounded by the other error sources (wind, bullet imbalance, parallax, sight).

5. Robustness to higher modes. Modes 2 and 3, of significantly higher frequencies (cf. the previous table), produce a fast oscillation superposed on $\theta(L, t)$. If they are poorly damped and significantly excited, they create a residual variability (*flyers*) even after tuning. A muzzle mass distinct from an integer harmonic of mode 1 (anti-resonance) helps filter them out, which motivates the “adjustable” double-ring designs.

6. Range of validity: centerfire vs rimfire. A caveat of intellectual honesty, stated by Kolbe himself [8], deserves to be reported. The *principle* of positive compensation (Sections 1 and 5) is general and was first established for rimfire. The *impulsive recoil-moment excitation model* adopted here (and in Kolbe’s simulator), however, is chiefly relevant to *centerfire* calibres: there the pressure–time curve is long (~ 1 ms, peak ~ 350 MPa $\approx 50,000$ psi, .308 Win-like profile) and the recoil high, so that the recoil moment does dominate the onset of vibration. For the .22 LR, the pressure pulse is so brief that, according to Kolbe, the recoil moment alone reproduces the observed vibrations poorly: *other* excitation sources (rifling engagement, the moving projectile’s weight effect, assembly play) take a comparable share. Our model captures them partially — this is the very purpose of contributions (b) and (c) of Section 3.4, absent from Kolbe’s original simulator — but the rimfire numbers given in this document should be read as *calibrated orders of magnitude*, not absolute predictions. Direct experimental validation (Section 5.5) remains the reference for rimfire.

7. A planar model: vertical vibration only. A second caveat of intellectual honesty concerns the dimensionality of the model. The whole preceding development is *planar*: the Euler–Bernoulli beam (2) is solved in a single plane and describes only the *vertical* component of the muzzle oscillation. Nothing, however, forces the barrel to vibrate only vertically: in reality the muzzle traces a *two-dimensional orbit* (an elliptical “whip”), and its horizontal component — already noted as uncompensated in point 4 — is corrected by no positive-compensation mechanism. It adds to the residual dispersion and sets a floor on the achievable precision. If the vertical model nonetheless remains relevant to first order, it is because *two effects break the symmetry* and privilege the vertical plane: *gravity*, which imposes a static sag and orients the dominant mode; and the *recoil geometry*, whose muzzle-rise couple (bore above the centre of gravity, Section 1) is essentially vertical. Above all, positive compensation (27) does not require the vibration to be *purely* vertical: it suffices that the vertical component of the muzzle angle correlate, in the right direction, with the exit time t_b (hence with the muzzle velocity). The horizontal motion adds noise without destroying this benefit, which the empirical success of tuners in .22 LR benchrest

confirms. The planar model is therefore a *first-order* model: useful and predictive, but which structurally underestimates the real dispersion.

8. A distance-specific tuning. The optimal angular velocity (28) is *proportional to the shooting distance*: $\dot{\theta}_{\text{out}}^* = -gD/(v_0^3 \tau_v) \propto D$. A tuner tuned for $D = 50$ m therefore realises equality (27) only at that distance; at another distance, the ballistic term (in $\partial h_{\text{bal}}/\partial v_0 \propto D^2$) and the angular term (in $\dot{\theta}_{\text{out}} D$) no longer cancel exactly and the compensation becomes *partial* — under-compensation beyond the tuning distance, over-compensation short of it —, the velocity dispersion gradually reappearing as vertical stringing. The lesson is that positive compensation does not *reduce* the muzzle-velocity dispersion of the lot (the velocity standard deviation is unchanged): it *masks* its vertical effect, and only near the tuning distance. A lot with a low velocity standard deviation therefore remains preferable, and a tuner is best verified at the competition distance. The phenomenon is nonetheless gradual: a setting established at short range retains part of its benefit at longer range as long as the intrinsic group remains smaller than the ballistic stringing it corrects.

9. A tuner is not always beneficial: Harral’s FEA study. A final caveat, an empirical one, comes from an independent finite-element simulation carried out by A. Harral [11] on a .22 LR benchrest rifle (“Esten’s” rifle). Computing, for the velocity range 1035–1075 fps, the on-target vertical spread with and without a muzzle mass, Harral obtains a counter-intuitive result: the *best* vertical group here is obtained *without* a tuner (0.091 inch, i.e. ≈ 2.3 mm), adding a mass *degrading* it monotonically — 0.122 inch (≈ 3.1 mm) with 4.9 oz (≈ 139 g), 0.183 inch (≈ 4.6 mm) with 16 oz (≈ 454 g). The interpretation is consistent with our formalism: this bare barrel already places, by its geometry (a flexible *reverse-taper* profile), the bullet exit t_b on a favourable *ascending* flank of $\theta(L, t)$; adding muzzle weight slows the kinematics (18) and drifts t_b toward a *descending* flank, where the combination becomes additive (dispersion *worsened*). Two practical lessons follow: (i) the tuner does not *create* compensation, it merely *shifts* t_b within the vibration cycle — if the barrel is already well placed, the best tuner position may be “no mass”, its role reducing to *re-tuning* after a change of lot or distance (point 3 and point 8); (ii) a *light* mass, with finer adjustment, is preferable to a heavy one that risks “overflying” the optimum from one notch to the next. Harral further confirms, in the same study, the distance-specific character (point 8): a setting made at 50 yd does not remain optimal at 100 yd.

A caveat on this warning. All of the above rests on a *single, purely numerical* source. Kolbe [8] — who did measure — considers that work unconfirmed: “His work has lacked the experimental confirmation needed to verify his computer modelling, however.” Harral publishes neither his rifle data (a single figure: 10.5 lb), nor his natural frequencies, nor his damping. The result remains plausible and consistent with the formalism, but must be read as an *unvalidated simulation prediction*, not as a measured fact.

10. In a vice or at the shoulder? The support condition is the mechanism. The most common practical question — should one tune with the rifle clamped in a vice, to “remove the human factor”? — calls for a blunt answer: *no*, and not for reasons of realism, but because a rigid vice *suppresses the very phenomenon one is trying to tune*. The driving moment (8) exists only because the rifle *recoils and pivots about its centre of gravity* (section 3.4); rigidly clamping the base cancels that term, leaving only elastic

bending of the tube, which produces *no* compensation at all. Kolbe’s own test rig was a clamp — but he points out that it was not rigid: “The relatively thin base plate flexed under recoil and allowed the barrel clamp to rotate backwards, resulting in an upwards vertical muzzle flip.” A truly rigid clamp would have shown nothing. For *benchrest*, the matter is settled: shooting off bags is, in Kolbe’s words, “a fair approximation” of the free recoil the model assumes. For *shouldered or prone shooting*, however, shouldering adds effective mass and constrains the rotation: since the amplitude scales as h_{cg}/m_r (9), a tune established in free recoil has no reason to remain optimal when shouldered. Kolbe is cautious (“if a small calibre rifle is gripped tightly or pulled hard into the shoulder then the recoil dynamics could be affected”) and did not measure it. On present knowledge the prudent rule is therefore: *tune in the position you shoot*, and re-check the tune if you change position or hold.

11. An unsuccessful recreation of Harral’s study. The analysis of point 9 was replayed within the FEM framework of this document — an Euler–Bernoulli beam of *variable section* reproducing Esten’s *reverse taper* contour, Newmark- β transient response, then Harral’s point-of-impact formula (muzzle projection + vertical velocity $\times$ time of flight – drop). The recreation *fails*, and the diagnosis illuminates the limits of the clamped model. Three causes were identified, two of them corrected.

(i) *Harral’s muzzle projection is not vibration but static sag* (corrected). His page states it explicitly: “Gravity is applied and the stock deforms as well as the barrel sag.” His $-1.32'' \rightarrow -2.42''$ describe a barrel sagging under the increasing weight of the tuner, before the shot is even fired. The script initially applied no gravity at all; it now applies the self-weight of barrel and tuner and starts from static equilibrium $\mathbf{U}_0 = \mathbf{K}_a^{-1} \mathbf{F}_{\text{grav}}$, giving $-1.13/-1.41/-1.61/-2.02''$ against Harral’s $-1.32/-1.72/-1.96/-2.42''$: the same trend and the same order of magnitude *with no fitting whatsoever*, the residual gap corresponding to the stock, which Harral models and this script ignores.

(ii) *The calibration compared a dynamic amplitude with a static number* (corrected). Scaling the peak of the oscillation to $\approx 1.5''$ — that is, to Harral’s *static* sag — inflated the amplitude by roughly an order of magnitude. With the static sag now produced by gravity, that scaling has gone; but the consequence must be drawn: *nothing in Harral’s published data constrains the dynamic amplitude any longer*. It has become an acknowledged free parameter.

(iii) *The boundary conditions describe a different system* (the decisive cause, unresolved). Harral models the entire rifle, free to recoil and rotate: “The stock contacts the simulated sandbag rests with zero friction in the calculation. The rests are fixed in space.” This is a *frictionless unilateral support* — neither a clamp nor a free-floating rifle. The clamped model, by contrast, suppresses all whole-body rotation, precisely the mechanism from which positive compensation arises (point 10). Only elastic bending remains: the 35 Hz fundamental is out of play and the muzzle angle oscillates at ~ 400 Hz, so that t_b falls on a steep slope of this rapid oscillation. The value read there is not a compensation but a sample taken at an arbitrary instant, and the exit cannot be shifted to recover the phase: at 315 m/s over 63 cm, the absolute floor is 2.0 ms.

Adjusting the amplitude corrects nothing, as the addition of gravity demonstrated. Before correction, the computed dispersion *decreased* as the muzzle was loaded ($0.148''$ bare $\rightarrow 0.085''$ at 16 oz) where Harral reports it *increasing* ($0.091'' \rightarrow 0.183''$): the opposite physical conclusion. Once the amplitude was brought back to a realistic value and the internal ballistics corrected (burnout profile, $t_b = 2.69$ ms), the inversion disappears and

the dispersion clusters around the *uncompensated* value — 0.159/0.168/0.170/0.169" for the four masses, all close to 0.173", with no trace of Harral's structure. The system being linear, this was predictable: what is missing is not a scale factor but the *rigid-body rotation* of the rifle under recoil. It is therefore a modelling gap, not an implementation error — a gap since closed by the free-rifle model (section 5.5).

Methodological caveats. The "rigid muzzle" line (0.173") recovered to the hundredth is not a validation: it is obtained by subtracting two drop values Harral publishes himself, with no simulation involved. Nor can his modal analysis be reproduced, since *he publishes no natural frequencies*. Finally, a trap awaits anyone replaying the exercise: in Harral's work, table and charts do not share the same reference — "The sag due to gravity was in the calculations, but subtracted out for the chart." The static sag is removed from the *curves*, not from the *proj* column of the table.

12. An excitation assumed identical on every shot. The model applies *exactly* the same recoil moment shot after shot: one and the same cartridge produces one and the same vibration, hence one and the same muzzle angle at exit. The assumption is convenient — it licenses speaking of *one* exit instant and *one* optimal setting — but it is false, and the discrepancy has been measured. Vaughn [5] instrumented the receiver ring with strain gauges: on a .270 Win, the peak moment is nominally 450 in-lb but *varies from 300 to 600 in-lb between shots with the same lot*, i.e. $\sim \pm 30\%$, the timing of the peak shifting as well. He attributes most of it to bolt-thrust variations and receiver asymmetries. On his unmodified rifle this *variability* — not the mean vibration — accounts on its own for ~ 0.8 inch (~ 20 mm) of dispersion at 100 yards. Three consequences for this document: a tuner set on *mean* behaviour still suffers that spread, which is a *floor* on a par with the ammunition's own dispersion; the narrow optimality window of section 5 is in fact *blurred*, since t_b and the amplitude do not repeat exactly; and the recoil moment is not the only source — Vaughn identifies *three* (recoil on the lug, bolt thrust on unevenly engaged lugs, receiver asymmetry) where (8) retains one, and removing the recoil component alone *does not* null the moment but exposes an opposite one. These figures are for a centerfire hunting rifle: their magnitude does not carry over to the match .22 LR, but the mechanism does.

Summary of epistemic status. The material gathered in this document is of unequal nature and benefits from being distinguished: *measured* — the principle of positive compensation and the 6.0 MOA/ms criterion (Kolbe, two concordant independent methods); *computed but unvalidated* — Harral's study, from which the warning of point 9 derives; *failed* — the recreation above, for want of whole-body rotation; *outside the model* — horizontal vibration (point 7), rimfire excitation beyond the recoil moment alone (point 6), and the shot-to-shot variability of that excitation, measured at $\pm 30\%$ (point 12).

7 The pre-charged pneumatic (PCP) case

Manufacturers of high-end air rifles offer muzzle *harmonic tuners*. These devices do have an effect, but through a mechanism distinct from the one developed in this document, and the distinction deserves to be stated precisely, as two schools of tuning coexist.

Positive compensation is largely out of play. The whole formalism above rests on *recoil*: it is recoil that rotates the rifle about its centre of gravity and imparts the driving moment (8). A PCP, however, is nearly recoilless, some models being designed as such. The dominant excitation term therefore collapses, and nothing guarantees any longer that the muzzle rises at the rate required by (28): that rate derived precisely from the recoil rotation.

The operative mechanism is the “stationary muzzle”.

School	Condition at exit	Effect	Requires recoil
Positive compensation (Kolbe, this document)	$\dot{\theta}$ maximal, positive (muzzle rising)	masks velocity spread; distance-specific	yes
Stationary muzzle (node)	$\dot{\theta} \approx 0$ (turning point)	makes the launch angle insensitive to jitter in t_b	no

The second condition does not depend on the excitation source. A PCP barrel does vibrate despite the absence of recoil, through the hammer strike on the valve, the air blast and the pellet’s passage (friction, engraving, moving-load effect) — this last term being already present in the model. A muzzle weight shifts the modal frequencies and moves t_b within this whip cycle.

At an extremum of θ , $\dot{\theta}$ vanishes but $\ddot{\theta}$ does not: the residual dispersion there is *second order* in Δt , varying as $\frac{1}{2}\ddot{\theta}\Delta t^2$. It is therefore minimised, not cancelled — the fundamental difference from positive compensation, which *uses* the first-order term $\dot{\theta}\Delta t$ to cancel the ballistic drop where the stationary muzzle *eliminates* it. Two confusions must be avoided: aiming at a maximum of $\dot{\theta}$ (the positive-compensation target, which requires recoil) instead of $\dot{\theta} \approx 0$; and aiming at an extremum of the *displacement* $y(t)$ rather than of the *angle* $\theta(t)$, when Kolbe measured that the muzzle’s translational velocity contributes negligibly to dispersion. In the real transient, θ and y peak at different instants.

Structure of a simulator. Counter-intuitively, the PCP moves *away* from Harral’s “free rifle” scheme. That elaborate scheme (free recoil, unilateral supports, rigid-body degrees of freedom) serves one purpose: to capture the whole-rifle rotation driven by recoil. As this term is negligible on a PCP, the shortcoming of the clamped model becomes immaterial: the clamped framework is in fact closer to its own assumptions for a PCP than for the .22 LR for which it was written. The bulk of the work therefore concerns the *excitation*: remove the recoil moment, add a hammer/valve impulse together with a realistic moving load, and target $\dot{\theta}(L, t_b) \approx 0$.

Results. This structure is implemented in `pcp_tuner.jl`. It exploits a property that makes the computation possible despite the unknown excitation: the system being linear, the *location* of the sweet spots ($\dot{\theta} = 0$) depends only on the unit response — mode shapes and exit time — not on the excitation amplitude. One can therefore predict *at which setting* the muzzle becomes stationary without measuring the hammer strike; only the *depth* of the gain remains out of reach.

The computation corrects a common assumption: one readily pictures a “comb” of closely spaced sweet spots. The whip does oscillate rapidly in time ($\sim 250\text{--}700$ Hz), but its frequency shifts by only $\sim 20\%$ when the tuner is loaded with 100 g, so the phase at t_b

sweeps only a fraction of a cycle. On a representative .22 airgun barrel, $\dot{\theta}(t_b)$ crosses zero only once within the realistic range. As in .22 LR (section 5.6), the effective adjustment is *position*: at fixed mass, $\dot{\theta}(t_b)$ crosses zero at a definite overhang — near 63 mm for 70 g, with a broad optimum (~ 18 mm). The mass shifts this sweet spot along the travel following a regular tuning curve (89 mm at 40 g, 63 mm at 70 g, 44 mm at 100 g): the heavier the weight, the closer the sweet spot to the muzzle.

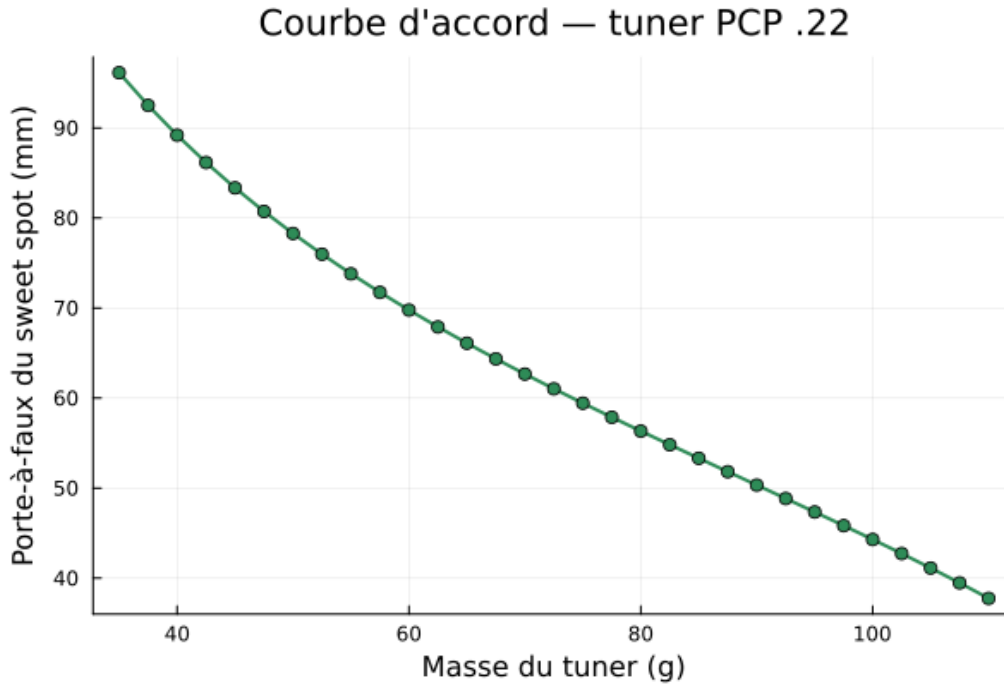


Figure 8: Tuning curve of a .22 PCP tuner: overhang of the sweet spot (stationary muzzle, $\dot{\theta} = 0$) as a function of the fixed mass. These values are specific to the simulated barrel; it is the *shape* of the relation that is robust, not the millimetre figures.

Reservations. The dominant accuracy variable on a PCP lies elsewhere: regulator consistency and pellet/velocity matching. A tuner compensates neither for a poorly set regulator nor for bad pellets, and on a well-regulated PCP little velocity spread remains to be masked. Many gains attributed to the tuner moreover stem from a pellet change tested at the same time — the same methodological pitfall as in .22 LR. Finally, the depth of the benefit depends on the excitation amplitude, which is not measured: the model gives the *where*, not the *how much*.

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